EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)

Topic D: Further Algebra and Functions

Sub topic D1: Sums and Products of Roots

Mark Scheme

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Topic **D: Further Algebra and Functions**

Sub topic **D1: Sums and Products of Roots**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

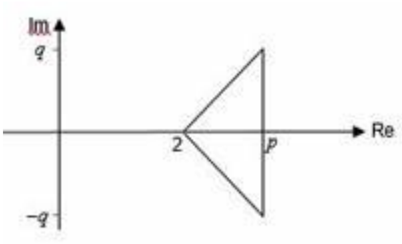
Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Writes the expression in terms of $\sum\alpha$ and $\sum\alpha\beta$ Award for correct expansion followed by use of $\sum\alpha^2 = (\sum\alpha)^2 - 2\sum\alpha\beta$	AO3.1a	M1	$(\alpha - \beta)^2 + (\gamma - \alpha)^2 + (\beta - \gamma)^2$ $= \alpha^2 - 2\alpha\beta + \beta^2 + \gamma^2 - 2\gamma\alpha + \alpha^2 + \beta^2 - 2\beta\gamma + \gamma^2$ $= 2\alpha^2 + 2\beta^2 + 2\gamma^2 - 2\alpha\beta - 2\gamma\alpha - 2\beta\gamma$ $= 2\sum\alpha^2 - 2\sum\alpha\beta$ $= 2((\sum\alpha)^2 - 2\sum\alpha\beta) - 2\sum\alpha\beta$
Substitutes $\pm m$ for $\sum\alpha$ **and** $\pm n$ for $\sum\alpha\beta$	AO1.1a	M1	$= 2(\sum\alpha)^2 - 6\sum\alpha\beta$ $= 2(-m)^2 - 6 \times n = 2m^2 - 6n$
Gives a reason for expression ≥ 0 Condone lack of reference to roots being real.	AO2.4	E1	But as α, β and γ are real then each of $(\alpha - \beta)^2$, $(\gamma - \alpha)^2$ and $(\beta - \gamma)^2$ must be non-negative. $\therefore (\alpha - \beta)^2 + (\gamma - \alpha)^2 + (\beta - \gamma)^2 \geq 0$
Completes fully correct proof to reach the required result. This mark is only available if all previous marks have been awarded. Lose this mark for sight of $\sum\alpha = m$	AO2.1	R1	$\therefore 2m^2 - 6n \geq 0$ $2m^2 \geq 6n$ $m^2 \geq 3n$ **AG**
			Total 4 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Writes β and γ in the form $p \pm q\text{i}$ (seen anywhere in the solution)	AO2.5	B1	Real coefficients $\Rightarrow \beta = p + q\text{i}$ and $\gamma = p - q\text{i}$
Uses "sum of the roots = −b/a" together with a conjugate pair to determine the real part (p) of β and γ	AO3.1a	M1	$\alpha + \beta + \gamma = 8$ $\Rightarrow 2 + p + q\text{i} + p - q\text{i} = 8$ $\Rightarrow 2 + 2p = 8$ $\Rightarrow p = 3$
Uses '(their p)' − 2 and the area of the triangle on an	AO3.1a	M1	$(p - 2)q = 8$

Argand diagram to determine the imaginary parts of β and γ			$\Rightarrow q = 8$ 
Uses a correct method to find the value of c or d using 'their' values of $p \pm q\mathrm{i}$	AO1.1a	M1	$\beta = \; 3 + 8\mathrm{i}$ and $\gamma = 3 - 8\mathrm{i}$
Obtains correct values for c and d. CAO	AO1.1b	A1	$d = -\alpha\beta\gamma = -146$ $c = \sum \alpha\beta = 85$
			Total 5 marks

Q3.

(a) $\alpha + \beta = -2$

B1

$\alpha\beta = -5$

B1

2

(b) $\alpha^2 + \beta^2 = (\alpha^2 + \beta^2)^2 - 2\alpha\beta = (-2)^2 - 2(-5)$

OE Using correct identity for $\alpha^2 + \beta^2$ with ft or correct substitution

M1

$= 14$

CSO A0 if $\alpha + \beta$ has wrong sign

A1

2

(c) $\alpha^3\beta + \alpha\beta^3 = \alpha\beta\,(\alpha^2 + \beta^2)$

PI Seen at least once in part (c).

OE eg $\alpha^3\beta + \alpha\beta^3 = \alpha\beta[(\alpha + \beta)^2 - 2\alpha\beta]$

M1

S(um)$= \alpha^3\beta + \alpha\beta^3 + 2 = (-5)(14) + 2 = -68$

Correct or ft c's $\alpha\beta$ × c's [answer (b)] + 2

A1F

P(roduct) = $(\alpha\beta)^4 + \alpha^3\beta + \alpha\beta^3 + 1 = (-5)^4 + (-5)(14) + 1 = 556$

Correct or

ft [c's $\alpha\beta$]^4 + c's $\alpha\beta$ × c's [answer (b)] + 1

A1F

$x - Sx + P$ (=0)

Using correct general form of LHS of eqn ***with*** *ft substitution of c's S and P values.*

M1

Eqn.: $x^2 + 68x + 556 = 0$

CSO ACF

A1 5

[9]

Q4.

(a) (i) $\alpha + \beta + \gamma = 5$

B1

$\alpha\beta\gamma = 4$

B1 2

(ii) $\alpha\beta\gamma^2 + \alpha\beta^2\gamma + \alpha^2\beta\gamma = \alpha\beta\gamma(\alpha + \beta + \gamma)$

M1

$= 5 \times 4 = 20$

FT their results from (a)(i)

A1√ 2

(b) (i) If α, β, γ are all real then $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 \geq 0$

Hence α, β, γ cannot all be real

argument must be sound

E1 1

(ii) $\alpha\beta + \beta\gamma + \gamma\alpha = k$

$\sum \alpha\beta = k$ *PI*

B1

$(\alpha\beta + \beta\gamma + \gamma\alpha)^2 = \sum\alpha^2\beta^2 + 2(\alpha\beta\gamma^2 + \alpha\beta^2\gamma + \alpha^2\beta\gamma)$

correct identity for $\left(\sum\alpha\beta\right)^2$

M1

$= -4 + 2(20)$

substituting their result from (a)(ii)

A1$\sqrt{}$

$k = \pm 6$

must see $k = \ldots$

A1 **cso**

4

[9]

Q5.

(a) $\alpha + \beta = -\frac{3}{2}$

OE

B1

$\alpha\beta = -3$

OE

B1

2

(b) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

Using correct identity for $\alpha^3 + \beta^3$ *in terms of* $\alpha + \beta$ *and* $\alpha\beta$.

M1

$= \left(-\frac{3}{2}\right)^3 - 3(-3)(-3/2)$

with ft / or correct substitution

A1F

$= -\frac{27}{8} - \frac{27}{2} = -\frac{135}{8}$

CSO AG. Correct evaluation of each of $(-1.5)^3$ *and* $-3(-3)(-1.5)$ *must be seen before the printed answer is stated*

A1

3

(c) Sum $= \alpha + \frac{\alpha}{\beta^2} + \beta + \frac{\beta}{\alpha^2} = \alpha + \beta + \frac{\alpha^3 + \beta^3}{(\alpha\beta)^2} = -\frac{3}{2} + \frac{-135/8}{9}$

Writing $\alpha + \frac{\alpha}{\beta^2} + \beta + \frac{\beta}{\alpha^2}$ *in a suitable form with ft/or correct substitution*

M1

Sum = $-\frac{27}{8}$

PI OE exact value eg − 3.375 (A0 if $\alpha\beta$ *= 3 used to get* $(\alpha\beta)^2 = 9$*)*

A1

Product = $\alpha\beta + \frac{\beta}{\alpha} + \frac{\alpha}{\beta} + \frac{1}{\alpha\beta} = \alpha\beta + \frac{\alpha^2 + \beta^2}{\alpha\beta} + \frac{1}{\alpha\beta}$ (*)

Now $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $(= \frac{9}{4} + 6)$

(*) OE with correct identity for $\alpha^2 + \beta^2$ *used in (c). Subst of values not required but PI by correct value of Product*

M1

Product = $-3 - \frac{1}{3}\left(\frac{9}{4} + 6\right) - \frac{1}{3} = -\frac{73}{12}$

PI OE exact value

A1

$x^2 - Sx + P$ (= 0)

Using correct general form of LHS of eqn ***with*** *ft substitution of c's* S *and* P *values.*

M1

Eqn is $24x^2 + 81x - 146 = 0$

OE but integer coefficients and '= 0' needed

A1

6

[11]

Q6.

(a) (i) $(\alpha\beta\gamma =)$ $-37 + 36\text{i}$

B1

1

(ii) $(\beta\gamma =)$ $(-2 + 3\text{i})(1 + 2\text{i}) = -2 + 3\text{i} - 4\text{i} - 6$

correct unsimplified but must simplify i^2

M1

$(-8 - \mathrm{i})\,\alpha = -37 + 36\mathrm{i}$
$\Rightarrow (8 + \mathrm{i})\,\alpha = 37 - 36\mathrm{i}$
***AG** be convinced*

A1cso
2

(iii) $\Rightarrow \alpha = \dfrac{37-36\mathrm{i}}{8+\mathrm{i}} \times \dfrac{8-\mathrm{i}}{8-\mathrm{i}}$

M1

$= \dfrac{296-37\mathrm{i}-288\mathrm{i}-36}{65}$
correct unsimplified

A1

$= \dfrac{260-325\mathrm{i}}{65} = 4 - 5\mathrm{i}$

A1cao
3

Alternative:
$(8 + i)(m + n\,i) = 37 - 36i$
$8m - n = 37;\ m + 8n = -36$

(M1)

either $m = 4$ or $n = -5$

(A1)

$\alpha = 4 - 5i$

(A1)
(3)

(b) $\alpha + \beta + \gamma = -p$
$-2 + 3\mathrm{i} + 1 + 2\mathrm{i} + 4 - 5\mathrm{i} = 3$
$(\Rightarrow p =)\ -3$

B1
1

(c) $\alpha\beta + \beta\gamma + \gamma\alpha = q$
$q = \sum\alpha\beta$ and attempt to evaluate three products FT "their" α

$(7 + 22\mathrm{i}) + (-8 - \mathrm{i}) + (14 + 3\mathrm{i}) = q$

M1

$q = 13 + 24\mathrm{i}$

A1cao
2

[9]

EXAM PAPERS PRACTICE

Q7.

(a) $\alpha + \beta = -\frac{7}{2}$

B1

$\alpha\beta = 4$

B1

(b) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{7}{2}\right)^2 - 2(4)$

Using correct identity with ft or correct substitution

M1

$= \frac{49}{4} - 8 = \frac{17}{4}$

CSO AG. A0 if $\alpha + \beta$ has wrong sign

A1

2

(c) (Sum =)

$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{17/4}{16}\left(= \frac{17}{64}\right)$

Writing $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ in a correct suitable form with ft or correct substitution

M1

$= \frac{17}{64}$

ft wrong value for $\alpha\beta$

A1F

(Product =) $\frac{1}{(\alpha\beta)^2} = \frac{1}{16}\left(= \frac{4}{64}\right)$

ft wrong value for $\alpha\beta$

B1F

$x^2 - Sx + P$ (= 0)

*Using correct general form of LHS of eqn **with** ft substitution of c's S and P values. PI*

M1

Eqn is $64x^2 - 17x + 4 = 0$

CSO Integer coefficients and '= 0' needed

A1

5

[9]

Q8.

(a) Use of $(\sum\alpha)^2 = \sum\alpha^2 + 2\sum\alpha\beta$

M1

AG

A1

2

(b) $p = 0$, $q = 5 + 6i$

B1,B1

2

(c) (i) Substitute 3i for z **or** use $3i\beta\gamma = -r$

allow for $3i\beta\gamma = r$

M1

$-27i + 15i - 18 + r = 0$ **or** $\beta\gamma = 5 + 6i + \alpha^2$

any form

A1

$r = 18 + 12i$

one error

A1F

3

(ii) Cubic is $(z - 3i)(z^2 + 3iz - 4 + 6i)$

or use of $\beta\gamma$ and $\beta + \gamma$

clearly shown

M1A1

2

(iii) $f(-2) = 0$ or equate imaginary parts

M1

$\beta = -2$, $\gamma = 2 - 3i$

correct answers no working and no check B1 only

A1,A1F

3

[12]

EXAM PAPERS PRACTICE

Q9.

Accept correct equivalent decimals in place of some / all fractions in the scheme

(a) $\alpha + \beta = \frac{7}{5}$ (= 1.4)

B1

$\alpha\beta = \frac{1}{5}$ (= 0.2)

B1

2

(b) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

OE eg $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\frac{1}{5}[7(\alpha+\beta) - 1 - 1]}{\alpha\beta}$ *scores M1 m1*

M1

$= \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{\left(\frac{7}{5}\right)^2 - 2\left(\frac{1}{5}\right)}{\frac{1}{5}}$

Correct expression for $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ *in terms of either* $(\alpha + \beta)$ *and* $\alpha\beta$ *or with numerical substitution of correct / c's values from (a)*

m1

$= \frac{\frac{49}{25} - 2\left(\frac{1}{5}\right)}{\frac{1}{5}} = \frac{\frac{49}{25} - \frac{2}{5}}{\frac{1}{5}} = \frac{\frac{39}{25}}{\frac{1}{5}} = \frac{39}{5}$

CSO AG must see some intermediate evaluation, must see one of the first three expressions

A0 if $\alpha + \beta$ *has wrong sign*

A1

3

(c) (Sum =) $\alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta} = \alpha + \beta + \frac{\alpha + \beta}{\alpha\beta}$

Writing $\alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta}$ *in a correct suitable form or with numerical values*

M1

$\left(=\frac{7}{5}+\frac{\frac{7}{5}}{\frac{1}{5}}\right)$

(Product =) $\alpha\beta + \frac{\alpha}{\beta}+\frac{\beta}{\alpha}+\frac{1}{\alpha\beta} = \frac{1}{5}+\frac{39}{5} + 5$

Correct expression for product into which substitution of numbers attempted for all terms, at least one either correct / correct ft

OE Both

M1

Sum = $\frac{42}{5}$, Product = 13

***SC** If B0 for $\alpha + \beta = -\frac{7}{5}$ in (a), and (c)*

$S = -\frac{42}{5}$ oe, $P = 13$ award this A1

A1

$x^2 - Sx + p$ (= 0)

*Using correct general form of LHS of equation **with** ft substitution of c's S and P values. PI. M0 if either $S = \alpha + \beta$ or $P = \alpha\beta$ values*

M1

Equation is $5x^2 - 42x + 65 = 0$

CSO Integer coefficients and '= 0' needed. Dependent on B1B1 in (a) and previous 4 marks in (c) scored

A1

5

[10]

Q10.

(a) (i) $\alpha + \beta + \gamma = 0$

B1

1

(ii) $\alpha\beta\gamma = -q$

B1

1

(b) $\alpha^3 + p\alpha + q = 0$

M1

$\sum\alpha^3 + p\sum\alpha + 3q = 0$

m1

$\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$
AG

A1

Alternative:

Use of $(\sum\alpha)^3 = (\sum\alpha^3) + 6\alpha\beta\gamma + 3(\sum\alpha \sum\alpha\beta - 3\alpha\beta\gamma)$

(M1)

Substitution of $\sum \alpha = 0$

(m1)

Result

(A1) 3

(c) (i) $\beta = 4 - 7\text{i}$, $\gamma = -8$

B1,B1 2

(ii) Attempt at either p or q

M1

$p = 1$

A1F

$q = 520$
ft incorrect roots provided p *and* q *are real*

A1F 3

(d) Replace z by $\frac{1}{z}$ in cubic equation
or $\sum\frac{1}{\alpha} = -\frac{p}{q}, \sum\frac{1}{\alpha\beta} = 0, \frac{1}{\alpha\beta\gamma} = -\frac{1}{q}$
ft on incorrect p *and / or* q

M1
A1F

$520z^3 + z^2 + 1 = 0$ coefficients must be integers
CAO

A1 3

[13]

Q11.

(a) $\alpha + \beta = 6$, $\alpha\beta = 18$

B1B1 **2**

(b) Sum of new roots = $6^2 - 2(18) = 0$

ft wrong value(s) in (a)

M1A1F

Product = $18^2 = 324$

ditto

B1F

Equation $x^2 + 324 = 0$

'= 0' needed here;

ft wrong value(s) for sum/product

A1F **4**

(c) α^2 and β^2 are $\pm 18\text{i}$

B1 **1**

[7]

Q12.

(a) $(1 + \text{i})^2 = 2\text{i}$ or $(1 + \text{i}) = \sqrt{2}e^{\frac{\pi i}{4}}$

B1

$2\text{i}(1 + \text{i}) = 2\text{i} - 2$

AG

Alternative method:

$(1 + i)^3 = 1 + 3i^2 + 3i^3 + i^3$ *B1*

$= 2i - 2$ *B1*

B1 **2**

(b) (i) Substitute $z = 1 + \text{i}$

M1

Correct expansion

allow for correctly picking out either the real or the imaginary parts

A1

$k = -5$

A1

3

(ii) $\beta + \gamma = 5 + i - \alpha = 4$

AG

B1

1

(iii) $\alpha\beta\gamma = 5(1 + i)$

allow if sign error

M1

$\beta\gamma = 5$

ft incorrect k

A1F

$z^2 - 4z + 5 = 0$

M1

Use of formula or $(z - 2)^2 = -1$

No ft for real roots if error in k

A1F

$z = 2 \pm i$

A1F

5

NB allow marks for (b) in whatever order they appear

[11]

Q13.

(a) $\alpha + \beta = -\frac{3}{2}$, $\alpha\beta = \frac{3}{4}$

B1B1

2

(b) $\alpha^2 + \beta^2 = \left(-\frac{3}{2}\right)^2 - 2\left(\frac{3}{4}\right) = \frac{3}{4}$

AG; A0 if α + β has wrong sign

M1A1

2

(c) Sum = $2(\alpha + \beta) = -3$

ft wrong value for $\alpha + \beta$

B1F

Product = $10\alpha\beta - 3(\alpha^2 + \beta^2) = \frac{21}{4}$

ft wrong values

M1A1F

$x^2 - Sx + P$ (= 0)

Signs must be correct for the M1

M1

Eqn is $4x^2 + 12x + 21 = 0$

Integer coeffs and '= 0' needed

A1

5

[9]

Q14.

(a) (i) $\sum \alpha = 2$

B1

$\sum \alpha\beta = 0$

B1

2

(ii) $\sum \alpha^2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta$

Used. Watch $\sum \alpha = -2$ *(M1A0)*

M1

= 4

AG

A1

2

(iii) Clear explanation

eg α satisfies the cubic equation since it is a root.

Accept $z = \alpha$

E1

1

(iv) $\sum \alpha^3 = 2\sum \alpha^2 - 3k$

Or $\sum \alpha^3 = \left(\sum \alpha\right)^3 - 3\sum \alpha \sum \alpha\beta + 3\alpha\beta\gamma$

M1

$= 8 - 3k$

AG

A1

2

(b) (i) $\alpha^4 = 2\alpha^3 - k\alpha$

B1

$\sum \alpha^4 = 2\sum \alpha^3 - k\sum \alpha$

Or $\sum \alpha^4 = \left(\sum \alpha^2\right)^2 - 2\left(\sum \alpha\beta\right)^2 + 4\alpha\beta\gamma \sum \alpha$

M1

$= 2\,(8 - 3k) - 2k$

ft on $\sum \alpha = -2$

A1

$k = 2$

AG

A1

4

(ii) $\sum \alpha^5 = 2\sum \alpha^4 - k\sum \alpha^2$

M1

Substitution of values

A1

$= -8$

A1

3

[14]

Q15.

(a) $\alpha + \beta = 2,\ \alpha\beta = \frac{1}{3}$

B1B1

2

(b) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

or other appropriate formula

M1

$\ldots = 8 - 3\left(\frac{1}{3}\right)(2) = 6$

m1 for substn of numerical values;
A1 for result shown (AG)

m1A1

3

(c) Sum of roots = $\dfrac{\alpha^3 + \beta^3}{\alpha\beta}$

M1

$\ldots = \dfrac{6}{1/3} = 18$

ft wrong value for αβ

A1F

Product = $\alpha\beta = \frac{1}{3}$

ditto

B1F

Equation is $3x^2 - 54x + 1 = 0$

Integer coeffs and "= 0" needed;
ft wrong sum and/or product

A1F

4

[9]

Q16.

(a) (i) $\beta = 2 - 2\sqrt{3}\,\mathrm{i}$

B1

1

(ii) $\alpha\beta\gamma = -8$

Allow for +8 but not ±16

M1

$\alpha\beta = 16$

B1

$y = -\frac{1}{2}$

A1 **3**

(iii) Either $\frac{-p}{2} = \alpha + \beta + \gamma$

or $\frac{q}{2} = \alpha\beta + \beta\gamma + \gamma\alpha$

SC *if failure to divide by 2 throughout,*
allow M1A1 for either p or q correct ft

M1

$p = -7, \quad q = 28$

ft incorrect γ

A1F,A1F **3**

Alternative to (a)(ii) and (a)(iii)

$(z^2 - 4z + 16)(az + b)$

(M1)

$\alpha\beta = 16$

(B1)

$a = 2, b = +1, \gamma = -\frac{1}{2}$

(A1)

Equating coefficients

(M1)

$p = -7$

(A1F)

$q = 28$

(A1F)

(b) (i) $r = 4, \theta = \frac{\pi}{3}$

B1,B1 **2**

(ii) $\left(2+2\sqrt{3}\,i\right)^n = \left(4e^{\frac{\pi i}{3}}\right)^n$

M1

$= 4^n\left(\cos\frac{n\pi}{3} + i\sin\frac{n\pi}{3}\right)$

AG

A1

2

(iii) $\left(2-2\sqrt{3}\,i\right)^n = 4^n\left(\cos\frac{n\pi}{3} - i\sin\frac{n\pi}{3}\right)$

B1

$\alpha^n + \beta^n + \gamma^n = 4^n\left(\cos\frac{n\pi}{3} + i\sin\frac{n\pi}{3}\right)$
$+ 4^n\left(\cos\frac{n\pi}{3} - i\sin\frac{n\pi}{3}\right) + \left(-\frac{1}{2}\right)^n$

M1

$= 2^{2n+1}\cos\frac{n\pi}{3} + \left(-\frac{1}{2}\right)^n$

AG

A1

3

[14]

Q17.

(a) $\alpha + \beta = 4$, $\alpha\beta = 10$

B1,B1

2

(b) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta}$

M1

$= \frac{4}{10} = \frac{2}{5}$

convincingly shown (AG)

A1

2

(c) Sum of roots = ($\alpha + \beta$) + 2(ans to (b))

M1

$= 4\frac{4}{5}$

ft wrong value for $\alpha + \beta$

A1F

Product = $\alpha\beta + 4 + \frac{4}{\alpha\beta}$

M1 for attempt to expand product (at least two terms correct)

M1A1

$= 14\frac{2}{5}$

ft wrong value for $\alpha\beta$

A1F

Equation is $5x^2 - 24x + 72 = 0$

integer coeffs and '= 0' needed here;
ft one numerical error

A1F

6

[10]

Q18.

(a) $\alpha + \beta + \gamma = 2$

B1

1

(b) (i) α is a root and so satisfies the equation

E1

1

(ii) $\sum \alpha^3 - 2\sum \alpha^2 + p\sum \alpha + 30 = 0$

M1A1

Substitution for $\sum \alpha^3$ and $\sum \alpha$

m1

$\sum \alpha^2 = p + 13$

AG

A1

4

(iii) $\left(\sum \alpha\right)^2 = \sum \alpha^2 + 2\sum \alpha\beta$ used

do not allow this M mark if used in (b)(ii)

M1

$p = -3$

AG

A1

2

(c) (i) f(–2) = 0

M1

$\alpha = -2$

A1

2

(ii) $(z + 2)(z^2 - 4z + 5) = 0$

For attempting to find quadratic factor

M1

$z = \dfrac{4 \pm \sqrt{-4}}{2}$

Use of formula or completing the square
m0 if roots are not complex

m1

$= 2 \pm i$

CAO

A1

3

[13]

Q19.

(a) Use of $\left(\sum \alpha\right)^2 = \sum \alpha^2 + 2\sum \alpha\beta$

M1

$1 = -5 + 2\sum \alpha\beta$

A1

$\sum \alpha\beta = 3$

AG

A1

3

(b) $1(-5-3) = -23 - 3\alpha\beta\gamma$

For use of identity

M1

$\alpha\beta\gamma = -5$

A1

2

(c) $z^3 - z^2 + 3z + 5 = 0$

M1

For correct signs and "= 0"

A1F

2

(d) $\alpha^2 + \beta^2 + \gamma^2 < 0 \Rightarrow$ non real roots

B1

Coefficients real $\Rightarrow$ conjugate pair

B1

2

(e) $f(-1) = 0 \Rightarrow z + 1$ is a factor

M1A1

$(z + 1)(z^2 - 2z + 5) = 0$

A1

$z = -1, 1 \pm 2i$

A1

4

[13]

Q20.

(a) $\alpha + \beta = -\frac{1}{2}, \alpha\beta = -4$

B1B1

2

(b) $\alpha^2 + \beta^2 = (-\frac{1}{2})^2 - 2(-4) = 8\frac{1}{4}$

M1 for substituting in correct formula;
ft wrong answer(s) in (a)

M1A1F

2

(c) Sum of roots = $4(8\frac{1}{4}) = 33$

ft wrong answer in (b)

B1F

Product = $16(\alpha\beta)^2 = 256$

ft wrong answer in (a)

B1F

Equation is $x^2 - 33x + 256 = 0$

ft wrong sum and/or product;
allow '$p = -33$, $q = 256$';
condone omission of '= 0'

B1F

3

[7]

Q21.

(a) 2 + 3i

B1

1

(b) (i) $\alpha\beta = 13$

B1

1

(ii) $\alpha\beta + \beta\gamma + \gamma\alpha = 25$

M1A0 for –25 (no ft)

M1

$\gamma(\alpha + \beta) = 12$

A1F

$\gamma = 3$

ft error in $\alpha\beta$

A1F

3

Alternative

Attempt at $(z - 2 + 3\text{i})(z - 2 - 3\text{i})$

(M1)

$z^2 - 4z + 13$

(A1)

cubic is $(z^2 - 4z + 13)(z - 3)$ $\therefore$ $\gamma = 3$

(A1)

(3)

(iii) $p = -\sum \alpha = -7$

M1 for a correct method for either p or q

M1A1F

$q = -\alpha\beta\gamma = -39$

ft from previous errors
p and q must be real
for sign errors in p and q allow M1 but A0

A1F

3

Alternative
Multiply out or pick out coefficients

(M1)

$p = -7$, $q = -39$

(A1,A1)

(3)

[8]

Q22.

(a) (i) $\alpha + \beta = 2$, $\alpha\beta = 4$

B1B1

$\alpha^3 + \beta^3 = (2)^3 - 3(4)(2) = -16$

M1A1

$\alpha^3 \beta^3 = (4)^3 = 64$, hence result

convincingly shown (AG)

M1A1

6

(ii) Discriminant 0, so roots equal

or by factorisation

B1E1

2

(b) $x = \dfrac{2 \pm \sqrt{4 - 16}}{2}$

or by completing square

M1

$\ldots = 1 \pm \frac{1}{2}\mathrm{i}\sqrt{12}$

A1

2

(c) $\alpha, \beta = 1 \pm \mathrm{i}\sqrt{3}$

and $\alpha^3 = \beta^3$, hence result

E2

2

[12]

Q23.

(a) (i) $\sum \alpha = -\mathrm{i}$

B1

1

(ii) $\sum \alpha\beta = 3$

B1

1

(iii) $\alpha\beta\gamma = 1 + \mathrm{i}$

B1

1

(b) (i) $\sum \alpha^2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta$ used

Allow if sign error or 2 missing

M1

$= (-\mathrm{i})^2 - 2 \times 3$

A1F

$= -7$

ft errors in (a)

A1F

3

(ii) $\sum \alpha^2\beta^2 = \left(\sum \alpha\beta\right)^2 - 2\sum \alpha\beta \cdot \beta\gamma$

Allow if sign error in 2 missing

M1

$= \left(\sum \alpha\beta\right)^2 - 2\alpha\beta\gamma\sum \alpha$

A1

= 9 – 2(1 + i)(–i)

ft errors in (a)

A1F

= 7 + 2i

ft errors in (a)

A1F

4

(iii) $\alpha^2\beta^2\gamma^2 = (1 + i)^2 = 2i$

M1

ft sign error in $\alpha\beta\gamma$

A1F

2

(c) $z^3 + 7z^2 + (7 + 2i)z - 2i = 0$

Correct numbers in correct places

B1F

Correct signs

B1F

2

[14]

Q24.

(a) $\alpha + \beta = -1$, $\alpha\beta = 5$

B1B1

2

(b) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

with numbers substituted

M1

... = 1 – 10 = –9

ft sign error(s) in (a)

A1F

2

(c) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

M1

$\ldots = -\frac{9}{5}$

AG: A0 if $\alpha + \beta = 1$ used

A1

2

(d) Product of new roots is 1

PI by constant term 1 or 5

B1

Eqn is $5x^2 + 9x + 5 = 0$

ft wrong value for product

B1F

2

[8]

Q25.

(a) (i) $\alpha\beta\gamma = -18 + 12\text{i}$

accept – (18 – 12i)

B1

1

(ii) $\alpha + \beta + \gamma = 0$

B1

1

(b) (i) $\alpha = -2$

B1F

1

(ii) $\beta\gamma = \frac{\alpha\beta\gamma}{\alpha} = 9 - 6\text{i}$

ft sign errors in (a) or (b)(i) or slips such as miscopy

M1A1F

2

(iii) $q = \sum \alpha\beta = \alpha(\beta + \gamma) + \beta\gamma$

M1

$= -2 \times 2 + 9 - 6i$

ft incorrect $\beta\gamma$ or α

A1F

$= 5 - 6i$

A1F

3

(c) $\beta = ki$, $\gamma = 2 - ki$

B1

$ki(2 - ki) = 9 - 6i$

M1

$2k = -6$ $(k^2 = 9)$ $k = -3$

imaginary parts

m1

$\beta = -3i$, $\gamma = 2 + 3i$

A1

4

[12]

Q26.

(a) $\alpha + \beta = -2, \alpha\beta = \frac{3}{2}$

B1B1

2

(b) Use of expansion of $(\alpha + \beta)^2$

M1

$\alpha^2 + \beta^2 = (-2)^2 - 2\left(\frac{3}{2}\right) = 1$

*convincingly shown (**AG**);*
m1A0 if $\alpha + \beta = 2$ used

m1A1

3

(c) $\alpha^4 + \beta^4$ given in terms of $\alpha + \beta$, $\alpha\beta$ and/or $\alpha^2 + \beta^2$

M1A0 if num error made

M1A1

$\alpha^4 + \beta^4 = -\frac{7}{2}$

OE

A1

3

[8]

Q27.

(a) $-k^3\text{i} + 2(1-\text{i})(-k^2) + 32(1+\text{i}) = 0$

Any form

M1

Equate real and imaginary parts:

$-k^3 + 2k^2 + 32 = 0$

A1

$-2k^2 + 32 = 0$

A1

$k = \pm 4$

A1

$k = +4$

AG

E1

5

(b) Sum of roots is –2(1 – i)

Or $\alpha\beta\gamma$ = – (32 + 32i)

Must be correct for M1

M1

Third root 2 – 2i

A1√

2

[7]

Q28.

(a) $\alpha + \beta = \frac{1}{2}, \alpha\beta = 2$

B1B1

2

(b) $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}$

M1

$\ldots=\frac{\frac{1}{2}}{2}=\frac{1}{4}$

Convincingly shown (AG)

A1

2

(c) Sum of roots = 1

PI by term ± x; ft error(s) in (a)

B1F

Product of roots = $\frac{16}{\alpha\beta}=8$

ft wrong value of $\alpha\beta$

B1F

Equation is $x^2 - x + 8 = 0$

ft wrong sum/product; "= 0" needed

B1F

3

[7]

Q29.

(a) $\sum \alpha\beta = 6$

B1

1

(b) (i) Sum of squares < 0 $\therefore$ not all real

E1

Coefficients real $\therefore$ conjugate pair

E1

2

(ii) $\left(\sum \alpha\right)^2 = \sum \alpha^2 + 2\sum \alpha\beta$

A1 for numerical values inserted

M1A1

$\left(\sum \alpha\right)^2 = 0$

A1F

$p = 0$

cao

A1F

4

(c) (i) $-1 - 3i$ is a root

B1

Use of appropriate relationship eg $\sum \alpha = 0$

M0 if $\sum \alpha^2$ *used unless the root 2 is checked*

M1

Third root 2

incorrect p $\surd$

A1F

3

(ii) $q = -(-1 - 3i)(-1 + 3i)^2$

allow even if sign error

M1

$= -20$

ft incorrect 3rd root

A1F

2

[12]

Q30.

(a) (i) Full expansion of product

M1

Use of $i^2 = -1$

m1

$(2 + \sqrt{5}i)(\sqrt{5} - i) = 3\sqrt{5} + 3i$

$\sqrt{5}\sqrt{5}$ *= 5 must be used − Accept not fully simplified*

A1

3

(ii) $z^* = x - iy$ (= $\sqrt{5}$ + i)

M1

Hence result

Convincingly shown (AG)

A1

2

(b) (i) Other root is $\sqrt{5}$ + i

B1

1

(ii) Sum of roots is $2\sqrt{5}$

B1

Product is 6

M1A1

3

(iii) $p = -2\sqrt{5}$, $q = 6$

B1

ft wrong answers in (ii)

B1ft

2

[11]

Q31.

(a) $p = -4$

B1

$(\alpha + \beta + \gamma)^2 = \Sigma\alpha^2 + 2\Sigma\alpha\beta$

M1

$16 = 20 + 2\Sigma\alpha\beta$

A1

$\Sigma\alpha\beta = -2$

A1F

$q = -2$

A1F

5

(b) 3 − i is a root

B1

Third root is −2

B1F

$\alpha\beta\gamma = (3 + i)(3 - i)(-2)$

M1

$= -20$

Real $\alpha\beta\gamma$

A1F

$r = +20$

Real r

A1F

5

Alternative to (b)
Substitute 3 + i into equation

M1

$(3 + i)^2 = 8 + 6i$

B1

$(3 + i)^3 = 18 + 26i$

B1

$r = 20$

Provided r is real

A2,1,0

[10]

Q32.

(a) $\alpha + \beta = 2,\ \alpha\beta = \frac{2}{3}$

***SC** 1/2 for answers 6 and 2*

B1B1

2

(b) (i) $(\alpha + \beta)^3 = \alpha^3 + 3\alpha^2\beta + 3\alpha\beta^2 + \beta^3$

Accept unsimplified

B1 **1**

(ii) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

M1

Substitution of numerical values

m1

$\alpha^3 + \beta^3 = 4$

convincingly shown AG

A1 **3**

(c) $\alpha^3\beta^3 = \frac{8}{27}$

B1

Equation of form $px^2 \pm 4px + r = 0$

M1

Answer $27x^2 - 108x + 8 = 0$

ft wrong value for $\alpha^3\beta^3$

A1ft **3**

[9]

Q33.

(a) (i) $\alpha + \beta + \gamma = 4\mathrm{i}$

B1 **1**

(ii) $\alpha\beta\gamma = 4 - 2\mathrm{i}$

B1 **1**

(b) (i) $\alpha + \alpha = 4\mathrm{i}, \quad \alpha = 2\mathrm{i}$

AG

B1 **1**

(ii) $\beta\gamma = \frac{4-2i}{2i} = -2i - 1$

Some method must be shown, eg $\frac{2}{i} - 1$

M1

AG

A1

2

(iii) $q = \alpha\beta + \beta\gamma + \gamma\alpha$

M1

$= \alpha(\beta + \gamma) + \beta\gamma$

Or $\alpha^2 + \beta\gamma$ *, ie suitable grouping*

M1

$= 2i.2i - 2i - 1 = -2i - 5$

AG

A1

3

(c) Use of $\beta + \gamma = 2i$ and $\beta\gamma = -2i - 1$

M1

$z^2 - 2iz - (1 + 2i) = 0$

Elimination of say γ *to arrive at* $\beta^2 - 2i\beta - (1 + 2i) = 0$ *M1A0 unless also some reference to* γ *being a root*

AG

A1

2

(d) f $(-1) = 1 + 2i - 1 - 2i = 0$

For any correct method

M1

$\beta = -1$, $\gamma = 1 + 2i$

A1 for each answer

A1A1

3

[13]

Q34.

(a) $\alpha + \beta = 5$, $\alpha\beta = -2$

B1, B1 2

(b) $\alpha^2\beta + \alpha\beta^2 = (\alpha + \beta) = -10$
ft wrong values

M1A1ft 2

(c) $(\alpha^2\beta)(\alpha\beta^2) = (\alpha\beta)^3 = -8$
ft wrong values

M1A1ft

Equation is $x^2 + 10x - 8 = 0$
Dep on both M1s; ft wrong values;
Condone omission of "= 0"

A1ft 3

[7]

Q35.

(a) (i) $\alpha + \beta = 4$, $\alpha\beta = 13$

B1B1

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

M1

$\ldots = 4^2 - 26 = -10$
convincingly shown (AG)

A1 2

(iii) The square of a real number is positive (or zero)

E1

The sum of two such squares is positive (or zero)

E1 2

(b) (i) $(\alpha + i) + (\beta + i) = 4 + 2i$
ft wrong value in (a)(i)

B1F 1

(ii) $(\alpha + i)(\beta + i) = 12 + 4i$
ditto

M1A1F

2

(c) Correct coeff of x or constant term

Using c's answers in (b)

M1

$x^2 - (4 + 2i)\,x + (12 + 4i) = 0$

ft wrong answers in (b)

A1F

2

[11]

Examiner reports

Q1.

Only the most able students were able to make progress with this question. Of these, there were some very good solutions which clearly explained why the expression had to be non-negative. Most students indicated that $-m$ was equal to the sum of the roots, and that n was equal to the sum of the product pairs, but few were able to use these, along with the roots equation $\sum\alpha^2 = (\sum\alpha)^2 - 2\sum\alpha\beta$, to make progress. Many attempts were simply abandoned halfway.

Q3.

This question, which tested roots and coefficients of a quadratic equation, proved to be a good source of marks for many students. Generally parts (a) and (b) were answered correctly. In part (c) a large majority of students used the appropriate identity $\alpha^3\beta + \alpha\beta^3 = \alpha\beta(\alpha^2 + \alpha^2)$ to find the sum and products of the new roots correctly, although the error $(-5)^4 = -625$ was not a rarity in finding an incorrect product. Although not as common as in recent papers, the omission of ' = 0' from the final answer was seen.

Q4.

(a) (i) This question was answered much better than in recent years, with very few making sign errors in either the sum or the product of the roots.

(ii) Almost everyone realised how to factorise the expression and to substitute the numerical values from the previous parts.

(b) (i) In contrast, very few explanations were considered good enough to earn the mark in this part. A proof by contradiction was the best approach, reasoning that if the roots α, β and γ were all real then $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$ could not be negative and hence if $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = -4$ then α, β and γ could not possibly all be real.

(ii) Some careless arithmetic was seen in this part and others were unable to form a correct identity in which to substitute values. There were two values for k, but for some reason some students thought that k had to be positive.

Q5.

This question tested roots and coefficients of a quadratic equation. Almost all students wrote down the correct values in part (a). In part (b) a large proportion of students quoted and used the correct identity for $\alpha^3 + \beta^3$ but a significant number failed to score the final mark because they did not show sufficient evaluations before writing down the printed answer. With the answer given, examiners expected to see correct evaluations of each of $(-1.5)^3$ and $-3(-3)(-1.5)$ before the printed answer was stated. Average grade students generally picked up the method marks in part (c) but, in general, only the more able students could cope with the algebraic demands needed to obtain the correct quadratic equation. However, it was pleasing to see that there were fewer students than normal who missed off the ' = 0' from their quadratic equation.

Q6.

This question was answered much better than similar questions in recent years, with very few making sign errors in either the sum or the product of the roots.

(a) (i) Almost everyone wrote down the correct product of the roots as a complex number. Very few forgot the associated minus sign.

(ii) The printed answer clearly helped with the product (– 2 + 3i)(1 + 2i) causing few problems. No doubt many did this calculation on a calculator and so in future questions involving algebra rather than simply numbers might be set.

(iii) Some used simultaneous equations to find the real and imaginary parts of the complex number, α. Others wrote $\alpha = \frac{37-3\mathrm{i}}{8+\mathrm{i}}$ and many were able to write this in the correct form using a calculator. It had been expected that students would have shown their working when evaluating $\frac{37-3\mathrm{i}}{8+\mathrm{i}} \times \frac{8-\mathrm{i}}{8-\mathrm{i}}$. The correct answer was obtained by the majority of students.

(b) Those who had the correct value of α were usually successful in finding the value of p. A small minority wrote p = 3, sometimes after obtaining $-p$ = 3 .

(c) Some careless arithmetic was seen in this part but most students found the correct value of q. The majority found the sum of three separate products and a few realised that they could use the value of $\beta\gamma$ found earlier and evaluated an expression of the form $\beta\gamma + \alpha(\beta + \gamma)$, which was slightly more efficient.

Q7.

This opening question, which tested roots and coefficients of a quadratic equation, proved to be a good source of marks for almost all candidates.

With the answer given in part (b), examiners expected candidates to show some evaluation between '$(-3.5)^2 - 2(4)$' and the printed answer $\frac{17}{4}$.

In part (c), where a quadratic equation was asked for, most candidates applied a correct method but a number of solutions were seen which ended without an equation, the '= 0' having been omitted.

Q8.

This question proved to be the most popular and probably the question on which candidates could secure the highest marks. Parts (a) and (b) were well done, as was part (c)(i). If an error occurred in part(c)(i), it was usually either the cube of 3i written down as -9i in the case where a candidate had substituted 3i for z in the cubic equation, or an error of sign in equating r to the product of the roots in the case when the roots of the cubic were used to find r. Solutions to part (c)(ii)were also pleasing, with many candidates showing that they needed to consider $\beta + \gamma$ as well as $\beta\gamma$ to show that β and γ were the roots of the quadratic equation. However a few thought that it was sufficient to show that one or other of $\beta + \gamma$ or $\beta\gamma$ satisfied the quadratic equation to establish the result. In part (c)(iii) many solutions were spoilt by errors of sign. For instance candidates who arrived at $\beta^2 = 4$, often just assumed that β was equal to 2.

Q9.

This opening question which tested roots and coefficients of a quadratic equation proved to be a good source of marks for almost all students. As indicated in the previous report for a similar question in this unit, with the answer given in part (b), examiners expected students to show more than just a full substitution followed by the printed answer. Many students reached the correct result $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\left(\frac{7}{5}\right)^2-2\left(\frac{1}{5}\right)}{1/5}$ and so scored the first two marks but a significant minority then just equated this to the printed value. Examiners expected to see at least $\left(\frac{7}{5}\right)^2$ evaluated correctly before the printed answer for the award of the final mark. In part (c), other than arithmetic errors, the most common loss of a mark was a failure to give an equation, the '= 0' having been omitted.

Q10.

Part (a) was invariably correctly quoted. However, in part (b) responses fell into several categories. The most successful students used the fact that α, β, and γ were each roots of the given cubic equation and therefore each satisfied that equation. Others quoted the formula for $\sum\alpha^3$ and, provided this quotation was correct, usually went on to secure full marks. A third category was those students who attempted to obtain $\sum\alpha^3$ by considering $(\sum\alpha)^3$ but these rarely succeeded, finding the algebra involved too complicated to handle. Some solutions ended at this juncture due to students not realising that a cubic equation with real coefficients must have a pair of complex conjugate roots if one root is complex. Part (c), apart from the odd slip of sign, was well done. In part (d), those students who replaced z by $\frac{1}{z}$ in the original cubic equation completed this part of the question quickly, but those who tried to find the sums and products of the reciprocals of α, β and γ usually ended up with at least one incorrect result.

Q11.

Most candidates started the paper in confident fashion, earning the first six marks with apparent ease, though some lost the sixth mark through failing to write '= 0' to complete their equation in part (b). Relatively few candidates realised that they were being tested on their knowledge of complex numbers in part (c), and even those who did sometimes failed to obtain the one mark on offer by misidentifying the two correct roots of their equation.

Q12.

Again part (a) was well done. Many candidates scored well in part (b) too, obtaining the results to parts (i) and (ii) in a variety of ways, although sign errors, especially in the product of the roots, did lead to a loss of marks. One common error was to think that, because $1 + \mathrm{i}$ was a root, $1 - \mathrm{i}$ also had to be a root, in spite of the fact that the coefficients of the cubic were complex.

Q13.

Almost all candidates earned the first four marks with no apparent difficulty. In part (c), there were small errors especially in finding the product of the roots of the required equation. In the forming of the equation itself, most candidates were aware of the need to reverse the sign of the sum of the roots to give the coefficient of x. Integer coefficients were usually obtained correctly, but, as mentioned above, many candidates failed to write the necessary '= 0' at the end.

Q14.

Apart from the weakest candidates, most scored full marks in parts (a)(i), (ii) and (iii), although in part (a)(iii) there was some confusion between roots and factors. Some candidates took the hint in part (a)(iii) in order to complete part (a)(iv) successfully, but others who tried to multiply out $(\alpha + \beta + \gamma)^3$ usually ended up in a morass of algebra. The same applied to part (b). Those candidates who were able to extend the result of part (a)(iii) to part (b) produced quick neat solutions. Some candidates also showed ingenuity in part (b) by using other methods which involved more algebra but managed to reach the correct results in the end. However, the solutions of the majority of candidates using longer algebraic methods usually petered out when uncertainty came as to how to handle their various expansions.

Q15.

Most candidates seemed to be very familiar with the techniques needed in this question. The formula for the sum of the cubes of the roots was either quoted confidently and correctly or worked out from first principles. Errors occurred mainly in part (c): algebraic errors in finding the sum or even the product of the roots of the required equation; errors in choosing which numerical values to substitute for $\alpha\beta$ or $\alpha + \beta$; and, very frequently, a failure to present the final answer in an acceptable form, with integer coefficients and the "= 0" at the end.

Q16.

This question provided a good source of marks for many candidates. The commonest error in part(a) was to write $\alpha\beta\gamma$ as –16, overlooking the fact that the coefficient of x^3 was not unity and, of course, leading to an incorrect value for γ. This error perpetuated in part (a)(ii) with $\alpha + \beta + \gamma$ written as p or $-p$ and the same for q.

Parts (b)(i) and (b)(ii) were generally well done, although it should be stated that when answers are printed, candidates are expected to provide sufficient detail to show clearly how their answers are arrived at. Part (b)(iii) was also quite well done and it was pleasing to note that in some cases where candidates had not arrived at $\gamma = -\frac{1}{2}$, they went back to part (a)(ii) to identify their error. Just occasionally in part (b)(iii), some candidates made blatent errors in their attempt to convert $4^n \cos\frac{n\pi}{3} + 4^n \cos\frac{n\pi}{3}$ into $2^{2n+1} \cos\frac{n\pi}{3}$.

Q17.

This was another well-answered question. The first two parts presented no difficulty to any reasonably competent candidate. In part (c) some candidates, faced with the task of finding the sum of the roots of the required equation, repeated the work done in part (b) rather than quoting the result obtained there. The expansion of the product of the new roots caused some unexpected difficulties, some candidates failing to deal properly with two terms which should have given them constant values. The final mark was often lost by a failure to observe the technical requirements spelt out in the question.

Q18.

Whilst parts (a) and (b)(i) were well done, few candidates were able to complete part (b)(ii) correctly through not taking note of the hint given in part (b)(i). Those candidates attempting to work out $(\alpha + \beta + \gamma)^3$ were inevitably doomed to failure. Part (b)(iii) was usually attempted by assuming the result of part (b)(ii). There were many correct solutions to part(c) although slips of sign often led to a solution with three real roots, contrary to the statement of part (c)(i).

Q19.

Parts (a), (b) and (c) of this question were well done apart from odd sign errors here and there. However, there was much woolly thinking in part (d). For instance, it was not uncommon to see statements such as 'the cubic equation has real coefficients so it must have one real root and a conjugate pair of non-real roots' or 'since $\alpha^2 + \beta^2 + \gamma^2 = -5$, two of α, β and γ must be non-real'. Part (e) was poorly answered: it just did not seem to occur to candidates to use the factor theorem to find the real root. Instead they tried to use the symmetric relations between the roots in order to find them. This in turn led to heavy algebra with final abandonment.

Q20.

Most candidates found this a very good introduction to the paper and gained full marks, or very nearly so. Errors in parts (a) and (b) were rare, and were nearly always sign errors rather than those caused by the use of an incorrect procedure.

In part (c) the most common mistake was to have a 4 instead of a 16 in the product of the roots of the required equation. Some candidates failed to see the connection with part (b) when finding the sum of the roots, but more often than not they still found the right value for this sum.

Q21.

Responses to this question were good, and the vast majority of candidates produced a completely correct solution. If errors did occur they were usually arithmetic, although occasionally p and q were given as $\sum \alpha$ and $\alpha\beta\gamma$ respectively with no consideration being given to their sign.

Q22.

This question proved to be an excellent source of marks for the majority of candidates, apart from the discriminating test provided by part (c). Many candidates seemed to be familiar with the techniques relating to the cubes of the roots of a quadratic equation, though some struggled to find the correct expression for the sum of the cubes of the roots, while others lost marks by not showing enough evidence in view of the fact that the required equation was printed on the question paper.

Parts (a)(ii) and (b) proved straightforward for most candidates, but only a few were able to give a clear explanation in part (c), where many candidates claimed that the original roots α and β must be equal.

Q23.

This question was probably the most popular question on this paper and certainly showed candidates well prepared to answer questions on this part of the specification. There were many fully correct solutions or correct apart from the odd sign error, the most common of which was to write down $(-i)^2$ as +1 instead of –1. If there was a major loss of marks, it was usually in the inability of a candidate to evaluate $\sum \alpha^2\beta^2$ and in this case the candidate started by considering $\left(\sum \alpha^2\right)^2$ only to find that the evaluation of $\sum \alpha^4$ posed a serious problem.

Q24.

The great majority of candidates showed a confident grasp of the algebra needed to deal with the sum of the squares of the roots of a quadratic equation, and answered all parts of this question efficiently. The only widespread loss of credit came in the very last part, where many candidates failed to give integer coefficients or else found a quadratic expression but without the necessary '= 0' to make it an equation.

Q25.

There were many correct solutions to parts (a) and (b) of this question. Errors, if they did occur, were largely errors of sign. However, in part (c), a good number of candidates did not use the hint given in the question to replace β by ki but instead called the two remaining roots β and $2 - \beta$. A quadratic equation in β ensued followed by the use of the quadratic formula and ended up with the square root of a complex number at which point solutions were abandoned.

Q26.

It was pleasing to note that almost all candidates were aware that the sum of the roots was −2 and not +2, and that they were able to tackle the sum of the squares of the roots correctly in part (b).

Part (c) was not so well answered. Relatively few candidates saw the short method based on the use of $(\alpha^2 + \beta^2)^2$. Of those who used the expansion of $(\alpha + \beta)^4$, many found the correct expansion but still had difficulty arranging the terms so that the appropriate substitutions could be made.

Q27.

Although this question was attempted by almost every candidate, there were few whose solutions presented the rigour required. Most substituted ki for z in the cubic equation, equated real parts and subsequently wrote $z^2 = 16$ so $z = 4$, not realising that $z = -4$ was also a root of $z^2 = 16$ and that imaginary parts had to be equated in order to reject the solution $z = -4$. Part (b) was not particularly well answered either. The most common errors were errors of sign in the use of $\alpha + \beta + \gamma$ or $\alpha\beta\gamma$, and those candidates using the product of the roots made extra work for themselves as they obtained a rational expression for γ which needed to be simplified. Some candidates thought that γ equalled –4i, the complex conjugate of α.

Q28.

Most candidates answered this question very well. Only a small number of candidates gave the wrong sign for the sum of the roots at the beginning of the question. Rather more made the equivalent mistake at the end of the question, giving the x term with the wrong sign. Another common mistake at the end of the question was to omit the 'equals zero', so that the equation asked for was not given as an equation at all.

Q29.

Whilst part (a) was usually correctly done, part (b)(i) was poorly answered. Some candidates were able to comment on the condition that as the sum of the squares of the roots was less than zero there would have to be complex roots, but few stated the conditions that the coefficients of the cubic equation were all real. The value of p in part (b)(ii) was very often correct but in part (c)(i) a very common error as to use $\sum \alpha^2 = -12$ in order to find the third root. This method led to $\alpha^2 = 4$ from which almost all candidates using this method wrote $\alpha = 2$ without even considering the possibility that α could equal –2. Part (c)(ii) was usually worked correctly although $\alpha\beta\gamma = +q$ appeared from time to time.

Q30.

Part (a)(i) was answered well by the majority of candidates, though the fourth term in the expansion often appeared as $-i^2$ instead of $-i^2\sqrt{5}$, and $-i^2$ was frequently simplified to –1 in the next line.

The next part, (a)(ii), tested the candidates' knowledge of technique to some extent. Those who understood what they were being asked to do invariably realised that the necessary hard work had already been done in part (a)(i), so that very little writing was needed to pick up two further marks. Those who thought they had to prove, rather than verify, the given statement were awarded one mark for showing a knowledge of the conjugate of z, but then spent much valuable time trying to solve the equation. There were three indications that this was not what the question required: the word 'Hence', the word 'verify', and the award of only two marks.

Part (b) of this question was generally well answered, though some candidates did not seem to be aware of the implications of the statement that the coefficients in the quadratic equation were real. The answers to part (b)(ii) were sometimes given in terms of p and q, and in part (b)(iii) many candidates equated p to the sum of the roots instead of to this number preceded by a minus sign.

Q31.

Candidates were also able to achieve good results on this question. In part (a), where errors occurred they were almost always errors of signs. For instance the value of p was given as 2 instead of –2 or the formula for $(\Sigma\alpha)^2$ was incorrectly quoted as $\Sigma\alpha^2 - 2\Sigma\alpha\beta$. There were fewer completely correct solutions to part (b) often due to inelegant methods of solution. The most successful candidates obtained the values of the other two roots and then worked out the product $\alpha\beta\gamma$. The main loss of marks using this method was to equate r to $\alpha\beta\gamma$ instead of $-\alpha\beta\gamma$. The other main method of approach to this part of the question was to substitute 3 + i into the cubic equation with the values of p and q already found in part (a). However any error in the values of p and q or in the substitution inevitably led to r having a complex value. Surprisingly this did not seem to worry the candidates in spite of the fact that the question stated that r was real.

Q32.

This opening question provided most candidates with a very good start to the paper, with full marks or nearly full marks earned. It was pleasing to see that almost all candidates gave the correct values in part (a), with no sign error for the sum of the roots, and that a very large proportion of candidates were careful to insert a minus sign and 'equals zero' in their equation in part (c). In part (b) (i) some candidates wrote down the correct expansion, while others worked it out laboriously from first principles. Some failed to understand the word ‘Expand’ and instead gave simply a numerical value. An even more unexpected mistake was to quote, or rather misquote, a formula for $\alpha^3 + \beta^3$ and write the absurd statement $(\alpha + \beta) = (\alpha + \beta) - 3\alpha\beta(\alpha + \beta)$. Luckily for these candidates, only one mark was lost and the rest of the question was often completed successfully.

Q33.

Apart from the occasional sign errors, part (a) was answered well. Where sign errors did occur there was some faking to establish the printed answers in part (b). Part (b) is an example of what was mentioned at the beginning of this report in that with all three answers being printed, sufficient working needed to be shown in order to obtain full credit. Whilst most candidates knew roughly what was required for part (c), few candidates could express their argument succinctly. A number of candidates attempted to divide the cubic equation by $z-2i$ with varying success. Probably the commonest method of approach in part (d) was to substitute β for z in the quadratic equation in z and then to equate real parts. Equating real parts led to $\beta^2 = 1$ from which a substantial number of candidates assumed that $\beta = 1$ instead of considering the imaginary parts of the equation as well.

Q34.

This question allowed most candidates to make a good confident start to the paper. Occasionally the values of $\alpha + \beta$ and $\alpha\beta$ were interchanged or one of them had the wrong sign. More seriously, in part (c), some candidates equated $(\alpha\beta)^3$ with $(\alpha + \beta)^3$ and often felt it necessary to expand $(\alpha + \beta)^3$, causing them a considerable loss of time. The mark scheme had only one mark for putting together the final equation. Hence, the distressingly common mistake of omitting the “= 0” at the end was not penalised on this occasion.

Q35.

There were many high-scoring attempts at this question, most candidates showing the necessary familiarity with the roots and coefficients of quadratic equations as well as with complex numbers. The explanation called for in part (a)(iii) often showed an awareness that real numbers cannot have negative squares, but failed to indicate the fact that $\alpha + \beta^2$ was not simply the square of α or of β. Some reference to the addition of the two squares was needed for full credit here.

In part (b) some candidates gave answers involving α and β rather than using the results they had obtained in part (a). In part (c) many candidates lost a mark by giving the wrong sign for the x term and/or omitting the ‘equals zero’ from their ‘equation’.